



Technical Service Bulletin

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QSK60 Construction 3000hp Rebuild Steel to FCD Power Cylinder

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

QSK60 CM2150 MRCS

Description of Change

All rebuild QSK60 CM2150 MCRS models running 3000hp uprate to be rebuilt with FCD (Ferrous Cast Ductile) power cylinder kit rather than the steel piston variant.

Reason for Change

There have been multiple reports of customers experiencing liner cavitation due to excessive secondary motion of Tier 2 steel pistons. Research has been conducted into whether a similar failure mode occurs on FCD, and data shows reduced secondary motion. This reduction in secondary motion results in less liner wall excitation; therefore, more margin against cavitation occurring.

Customer Communication

This change only applies to engines running the 3000Hp Tier 2 industrial configuration.

Service Instructions

This doesn't authorize field changeouts. This change should only be completed at rebuild or when full power cylinder replacement is required. FCD and steel pistons **must** not be mixed on an engine, a complete set is required.

Service Parts Availability

Service parts are available.

Table 1, Service Parts

Part Description	Existing Part Number	Obsolete	Superseded	New Part Number
Piston	5371866	No	No	3640796
Liner (Nominal)	3644545	No	No	4085917
Liner (0.25mm oversize)	3644548	No	No	3636998
Liner (0.50mm Oversize)	3644546	No	No	3636999
Liner (0.75mm Oversize)	3644547	No	No	3637000
Piston Top Ring	3650569	No	No	4910251
Piston Second Ring	2891821	No	No	4910253
Piston Oil Ring	4322612	No	No	3090450

Part Compatibility

This change requires a change of: Piston, liner and piston rings.

Part Identification

Part numbers are stamped on all components for identification.

Part Inventory Action

Please use steel piston parts with boron liner in applications that's are not relevant to the application for which this TSB is being released.

Document History

Date	Details
2020-2-27	Module Created

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